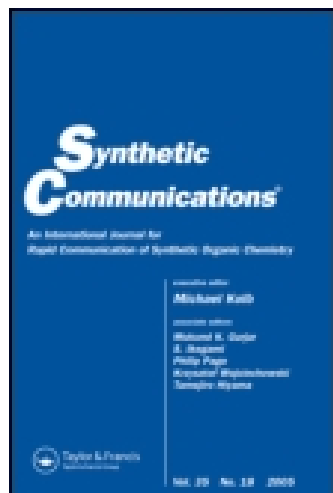




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Synthesis of Fluorinated Dimethyl Carbonates by Direct Fluorination

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ABSTRACT

Growing interest has been focused on the development of fluorinated solvents for lithium batteries, because they have a number of benefits such as excellent oxidation durability, wide liquidus ranges, and nonflammability. Fluorination of dimethyl carbonate (DMC) is carried out using molecular fluorine (15 vol.% F₂/N₂) at 5°C, and the fluorinated derivatives of the DMC are identified and characterized. The selectivity of monofluorinated dimethyl carbonate (MFDMC) is ca. 90% at early stage of the fluorination. The successive electrophilic substitution of a

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hydrogen with a fluorine is found to proceed for the direct fluorination of the DMC.

Key Words: Direct fluorination; Dimethyl carbonate; Fluorinated dimethyl carbonate; Lithium battery.

INTRODUCTION

There has been a growing demand for the development of novel solvents for the high-performance electrochemical devices. One of the strategies for finding novel solvents is the introduction of fluorine atoms into the solvent molecules. This is because the fluorinated solvents can improve various properties such as oxidation durability, liquidus ranges, and nonflammability. Since the molecular fluorine is very reactive, direct fluorination^[1] can afford various fluorinated compounds, which are hardly obtained by conventional methods, under solvent-free conditions. The fluorination can be performed safely at low temperatures using F₂ gas that is generated by electrolysis and diluted with an inert gas. Dimethyl carbonate (DMC) is one of the low-viscosity chain esters commonly used in electrochemical energy storage devices such as lithium batteries.^[2–4]

We previously reported direct fluorination of γ -butyrolactone (γ -BL)^[1] that is also commonly used in the electrochemical devices. The direct fluorination gives β -fluoro- γ -butyrolactone and γ -fluoro- γ -butyrolactone besides α -fluoro- γ -butyrolactone. Similar direct fluorination is applied to 1,2-dimethoxyethane (DME),^[5,6] propylene carbonate (PC),^[7] and ethylene carbonate (EC).^[7,8] No attempt has been made to conduct direct fluorination of acyclic esters, though perfluorodimethyl carbonate^[9,10] and *gem*-difluorinated DMC^[11] (*gem*-DFDMC in Sch. 1), where two fluorine atoms and an oxygen atom are attached to the same carbon atom, are synthesized by different methods. There are some reports of electrochemical fluorination of substrates such as *N*-containing carboxylic acids,^[12] amino alcohols,^[13] chain ethers,^[14,15] and cyclic ethers.^[16]

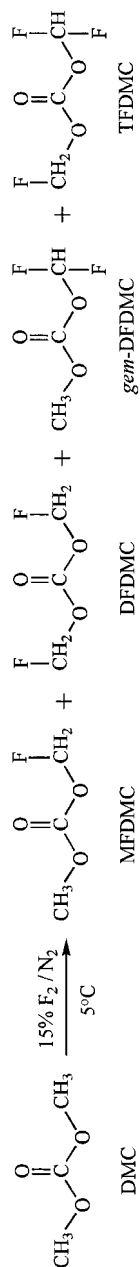
A fluorine atom is highly electronegative and shows a very low electronic polarizability and high enthalpy of ionization. Perfluorination or polyfluorination of solvents results in a lowering of their polarity. On the other hand, partial fluorination of solvents can lead to an increase of the polarity, though the amount of polarization depends on the position where the substitution takes place. It is desirable to develop partially fluorinated solvents for the high-performance electrochemical devices. We report here direct fluorination of DMC.

Figure 1 shows a gas chromatograph of the reaction mixture obtained after the introduction of F₂ gas for 24 hr. There appeared four peaks except for a peak due to DMC, a starting material. The result of GC-MS (CI) for



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Scheme 1.



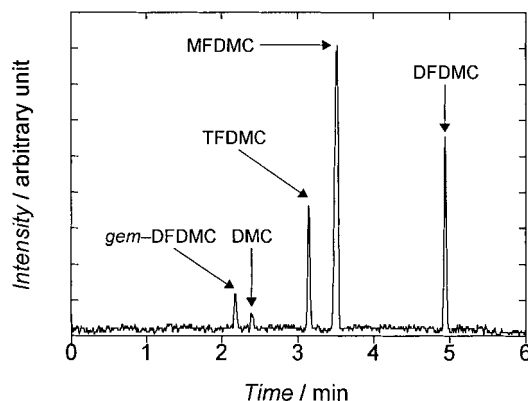


Figure 1. Gas chromatograph of the reaction mixture obtained after the introduction of F_2 gas for 24 hr.

the reaction mixture^a indicates that these peaks can be ascribed to molecular ions and that direct fluorination of DMC occurred without the fragmentation of the DMC chain.

A peak ascribable to the DMC, whose molecular weight is 90, was observed at the retention time of 2.3 min. A peak due to monofluorinated DMC (MFDMC), whose molecular weight is 108, appeared at 3.5 min. Two peaks attributable to difluorinated DMCs, whose molecular weights are 126, were detected at 2.1 and 4.9 min. According to the analysis of GC-MS (EI) spectra,^b the peak at 4.9 min can be ascribed to DFDMC, while the peak at 2.1 min can be accounted for *gem*-DFDMC. Furthermore, a peak responsible for trifluorinated DMC (TFDMC), whose molecular weight is 144, is observed at 3.1 min. The MFDMC purified by fractional distillation was also identified by techniques of elemental analysis^c and 1H , ^{13}C , and ^{19}F NMR spectroscopy.^d

^aGC-MS (CI) (m/z): DMC, 91 $[M + H]^+$; MFDMC, 109 $[M + H]^+$; DFDMC, 127 $[M + H]^+$; *gem*-DFDMC, 127 $[M + H]^+$; TFDMC, 145 $[M + H]^+$.

^bGC-MS (EI) (m/z): DMC, 90 $[M]^+$ (11.42), 59 $COOCH_3^+$ (100), 31 CH_3O^+ (73.14); MFDMC, 77 $COOCH_2F^+$ (4.33), 59 $COOCH_3^+$ (84.25), 33 CH_2F^+ (100); DFDMC, 77 $COOCH_2F^+$ (15.74), 33 CH_2F^+ (100); *gem*-DFDMC, 95 $COOCHF_2^+$ (2.38), 59 $COOCH_3^+$ (30), 51 CHF_2^+ (100), 31 CH_3O^+ (30.95); TFDMC, 33 CH_2F^+ (46.27), 51 CHF_2^+ (100), 77 $COOCH_2F^+$ (7.05), 95 $COOCHF_2^+$ (3.52).

^cAnal. Calcd for $C_3H_5FO_3$: C, 33.3%; H, 4.7%; F, 17.6%. Found: C, 33.4%; H, 3.9%; F, 17.5%.



The DFDMC and the *gem*-DFDMC are isomeric with each other. It is interesting to note that the retention time of the DFDMC (4.9 min) is considerably longer than that of the DMC (2.3 min), but that the retention time of the *gem*-DFDMC (2.1 min) is slightly shorter. The retention time reflects the boiling point of the solvent, which relates to the attractive forces between molecules. The boiling points of *gem*-DFDMC, DMC, TFDMC, MFDMC, and DFDMC were 85°C, 90°C, 100°C, 108°C, 120°C, respectively. The DFDMC molecule contains two carbon atoms, each of which is attached to a fluorine atom and an oxygen atom, whereas two fluorine atoms and an oxygen atom are bound to the same carbon atom in the *gem*-DFDMC molecule. A fluorine and an oxygen atom are highly electronegative and show very low electronic polarizability and high enthalpy of ionization. Accordingly, the polarity of the DFDMC can be substantially higher than that of the *gem*-DFDMC, whose electronic polarizability may be also small as compared to that of the DMC. It is expected that the drastic increase of the retention time (the boiling point) observed for the DFDMC can be ascribed to the dipole–dipole interaction, which includes the hydrogen bonding, as well as the dispersion or London interaction between molecules.

In general, as molecular weight increases, so too do the molecular size and surface area. With increasing them, the dispersion or London interactions between molecules increase.^[17] Therefore, more energy (a higher temperature) is required to separate molecules from one another. In spite of higher molecular weight of the TFDMC, the retention time of the TFDMC (3.1 min) was shorter than those of the MFDMC (3.5 min) and the DFDMC (4.9 min). Trifluorination of the DMC can induce a lowering of polarity as compared to the monofluorination and the difluorination. On the other hand, the retention time of the TFDMC was longer than that of the DMC (2.3 min), because the molecular weight and polarity of the TFDMC are higher than those of the DMC.

Figure 2 shows the variation of yields of the fluorinated derivatives of DMC with reaction time. Monofluorination of DMC preferentially proceeded, and the maximum yield of MFDMC (ca. 60%) was attained at 16 hr after the introduction of F₂ gas. The selectivity of the MFDMC was ca. 90% at early stage of the fluorination. The selectivity is defined as the ratio of the amount of the resulting MFDMC to that of the DMC consumed by the reaction. The yields and selectivity were estimated from the ratios of the areas of peaks observed on a gas chromatograph. The yields of two difluorinated DMCs

^{d1}H NMR (1% TMS/CDCl₃, 500.00 MHz): δ 4.23 (s, 3H), 6.08 (d, ²J_{HF} = 51.5 Hz, 2H). ¹³C NMR (1% TMS/CDCl₃, 125.65 MHz): δ 55.10, 96.97 (d, ¹J_{CF} = 219.0 Hz), 154.68. ⁹F NMR (2% CF₃COOD/D₂O, 470.40 MHz): δ -158.05 (t, ²J_{FH} = 51.5 Hz).



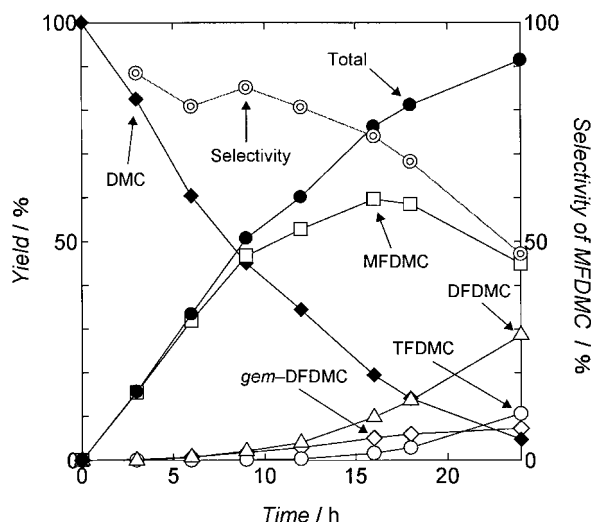


Figure 2. Yields of the fluorinated derivatives of DMC and selectivity of MFDMC as a function of reaction time.

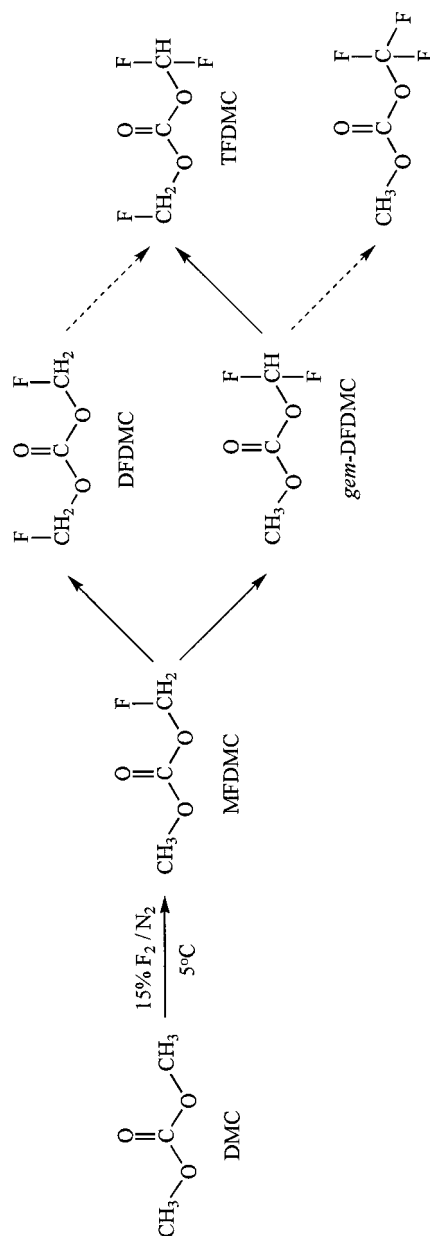
increased with the decrease of the yield and selectivity of MFDMC beyond 16 hr. No significant formation of these two difluorinated DMCs was observed at early stage. These findings indicate that the two difluorinated DMCs can result from the MFDMC. The yield of TFDMC was also extremely low at early stage. This yield exceeds that of *gem*-DFDMC after ca. 21 hr, while that of DFDMC continues to increase. Therefore, we can think that trifluorination of the DMC mainly occurs through the *gem*-DFDMC. On the other hand, no significant formation of another TFDMC, where three fluorine atoms are situated on the same carbon atom, was observed. The yield of the DFDMC was higher than that of the *gem*-DFDMC over an investigated period of time. It appears that a fluorine molecule acts as an electrophile and that electron withdrawing groups deactivate nearby C–H bonds toward the electrophilic attack.^[18,19] The process of the direct fluorination of the DMC is summarized in Sch. 2.

In conclusion, the selectivity of the MFDMC was ca. 90% at early stage of direct fluorination. The successive electrophilic substitution of a hydrogen with a fluorine was found to proceed for the direct fluorination of DMC. Difluorination of the DMC accompanies the consumption of the MFDMC. Tifluorination of the DMC mainly occurred through the *gem*-DFDMC, in which two fluorine atoms and an oxygen atom are attached to the same carbon atom. It is of interest to investigate the electrolytic properties of these



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Scheme 2.

fluorinated DMCs and to clarify the relationship between the successive substitution and the properties.

EXPERIMENTAL

Direct fluorination of DMC is depicted in Sch. 1. In a PFA vessel (500 mL) equipped with gas inlet and outlet 50.0 g (0.556 mol) of DMC was placed, and the atmosphere in the reactor was replaced with N₂ gas. F₂ gas generated by electrolysis of KF·2HF (FluorodecTM 30, Toyo Tanso Co. Ltd.) was diluted with N₂ gas to 15 vol.% and poured over the surface of DMC in the reactor at 22.4 mL min⁻¹ (6.00×10^{-2} mol hr⁻¹). The DMC solution was stirred and maintained at 5°C with cooling. The fluorinated derivatives of the DMC were identified and characterized by means of gas chromatography coupled with mass spectrometry (GC-MS analysis). The GC-MS analysis based on chemical ionization (CI)^a and electron ionization (EI)^b were performed on JEOL JMS-SX102A equipped with a TC-WAX column (0.25 mmφ × 30 m, df = 0.25 μm). Perfluorokerosene (PFK) was used as the internal standard. The temperature in the capillary column was increased at a rate of 10°C min⁻¹ from 50°C to 100°C. Although the capillary column was heated at a rate of 20°C min⁻¹ from 100°C to 200°C, no peaks were observed on the gas chromatograph. The boiling points of *gem*-DFDMC, DMC, TFDMC, MFDMC, and DFDMC were 85°C, 90°C, 100°C, 108°C, 120°C, respectively.

MFDMC was separated from the reaction mixture obtained after the introduction of F₂ gas for ca. 10 hr. Hydrogen fluoride, HF, can be formed as a by-product by the substitution. Since the fluorinated DMC solution was immiscible with water, the reaction mixture was washed with water and followed by washing aqueous solution containing Na₂CO₃ for removal of the HF. The fluorinated DMC solution was dehydrated with purified molecular sieve 4 A for ca. 24 hr. We have isolated the MFDMC from the mixture of the DMC and the fluorinated DMCs in a 45% yield using fractional distillation at atmospheric pressure. It should be noted that this yield is essentially the same as that shown in Fig. 2. That is to say, almost all of the MFDMC formed by the direct fluorination of the DMC was successfully separated. The purity of the MFDMC determined by means of GC (GC-1700, Shimadzu) was 99.7%. ¹H, ¹³C, and ¹⁹F NMR spectra of the purified MFDMC^d were recorded on a JEOL JNM-LA500 FT-NMR spectrometer.

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